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An asymptotic supnorm estimate for solutions of 1-D systems of convection–diffusion equations

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Abstract

We show how to use the L^p–L^q approach to obtain fundamental estimates for the spatial supnorm values of solutions $\boldsymbol{u}(x,t) = (u_1(x,t), \dots, u_m(x,t))$ to general systems of convection–diffusion equations of the form

$$\frac{\partial u_i}{\partial t} + \frac{\partial}{\partial x} f_i(x,t,u_1,\dots,u_m) + \frac{\partial}{\partial x} g_i(t,u_i) = \mu_i(t) \frac{\partial^2 u_i}{\partial x^2}, \quad 1 \le i \le m,$$

with initial data $\boldsymbol{u}(\cdot,0) \in L^{p_0}(\mathbb{R}) \cap L^\infty(\mathbb{R})$ for some $1 \le p_0 < \infty$, where $\mu_i(t) > 0$, given arbitrary $\boldsymbol{f} = (f_1,\dots,f_m)$, $\boldsymbol{g} = (g_1,\dots,g_m)$ such that $|\boldsymbol{f}(x,t,\boldsymbol{u})| \le B(t)|\boldsymbol{u}|$ for all $x \in \mathbb{R}$, $t \ge 0$, $\boldsymbol{u} \in \mathbb{R}^m$, where $B \in C^0([0,\infty[)$.

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1. Introduction

We derive some fundamental estimates for the supnorm values $\|\boldsymbol{u}(\cdot,t)\|_{L^\infty(\mathbb{R})}$ of solutions $\boldsymbol{u}(\cdot,t)=(u_1(\cdot,t),\ldots,u_m(\cdot,t))$ to general systems of convection–diffusion equations

$$\frac{\partial u_i}{\partial t}+\frac{\partial}{\partial x}\bigl\{f_i\bigl(x,t,\boldsymbol{u}(x,t)\bigr)+g_i\bigl(t,u_i(x,t)\bigr)\bigr\}=\mu_i(t)\frac{\partial^2 u_i}{\partial x^2},\quad 1\le i\le m, \tag{1.1a}$$

with initial data

$$\boldsymbol{u}(\cdot,0)\in L^{p_0}(\mathbb{R})\cap L^\infty(\mathbb{R}),\quad 1\le p_0<\infty, \tag{1.1b}$$

where[1] $\boldsymbol{f}=(f_1,\ldots,f_m)$, $\boldsymbol{g}=(g_1,\ldots,g_m)$ are arbitrary smooth[2] functions, with $\boldsymbol{f}$ satisfying

$$\bigl|\boldsymbol{f}(x,t,\boldsymbol{u})\bigr|\le B(t)|\boldsymbol{u}|\quad \forall x\in\mathbb{R},\ t\ge 0,\ \boldsymbol{u}\in\mathbb{R}^m \tag{1.2}$$

for some continuous $B(t)\ge 0$, where for definiteness $|\cdot|$ denotes the Euclidean norm,[3] and the viscosity coefficients $\mu_i\in C^0([0,\infty[)$, $1\le i\le m$, are all assumed positive. By *solution* to (1.1) in some time interval $[0,T_*[$, $0<T_*\le\infty$, we mean a function $\boldsymbol{u}=(u_1,\ldots,u_m):\mathbb{R}\times[0,T_*[\,\to\mathbb{R}^m$ which is bounded in each finite strip $S_T=\mathbb{R}\times[0,T]$, $0<T<T_*$ (i.e., $\boldsymbol{u}(\cdot,t)\in L^\infty_{\mathrm{loc}}([0,T_*[,L^\infty(\mathbb{R})^m))$, solves Eq. (1.1a) in the classical sense for $0<t<T_*$, and satisfies the condition (1.1b) in the sense that $u_i(\cdot,t)\to u_i(\cdot,0)$ in $L^1_{\mathrm{loc}}(\mathbb{R})$ as $t\to 0$ for each $1\le i\le m$. One can actually construct, for some $T>0$, a solution $\boldsymbol{u}(\cdot,t)\in C^0([0,T],L^{p_0}(\mathbb{R})^m)\cap L^\infty([0,T],L^\infty(\mathbb{R})^m)$, unique within this class, see e.g. [3,7,9,11,18,20] where the heat semigroup is used, or [1,8,12,14] for other methods. It then follows from Theorem 2.1 below that solutions can be continued for as long as they remain bounded, making estimates for $\|\boldsymbol{u}(\cdot,t)\|_{L^\infty(\mathbb{R})}$ the key ingredient to global existence results for systems like (1.1). Our main goal in this work is to show how the traditional L^p–L^q approach (see e.g. [2,4,17,18,21,22] and references therein) can be used to estimate $\|\boldsymbol{u}(\cdot,t)\|_{L^\infty(\mathbb{R})}$ in this case. In particular, it will be seen that, as it might be expected, all solutions to (1.1) turn out to be globally defined (i.e., $T_*=\infty$).

To have a feeling about the basic difficulties involved in estimating solution supnorm values under such general conditions, let us consider for the sake of illustration the simple scalar equation $u_t+(b(x,t)u)_x=\mu(t)u_{xx}$, with $b(x,t)$ smooth and bounded in some fixed strip S_T but otherwise arbitrary. Writing the equation as $u_t+b(x,t)u_x=\mu(t)u_{xx}-b_x(x,t)u$, we see that $|u(x,t)|$ can potentially increase in regions where $b_x(x,t)<0$, particularly if $-b_x(x,t)\gg 1$. How big can then $|u(x,t)|$ become on S_T? In the special event that $b_x(x,t)\ge 0$ everywhere, the answer is well known: all norms $\|u(\cdot,t)\|_{L^p(\mathbb{R})}$, $p_0\le p\le\infty$, decrease monotonically in time, and $\|u(\cdot,t)\|_{L^\infty(\mathbb{R})}=O(t^{-p_0/2})$, see e.g. [5,9,17,19,22]. Let us henceforth assume that $b_x\not\ge 0$. Now, as solution norms are still bounded on $[0,T]$, suppose that we have $\|u(\cdot,t)\|_{L^p(\mathbb{R})}\le K(T)$, $0\le t\le T$, for some finite $p\ge p_0$. This puts a limitation on the increase of supnorm values in this interval: if $|u(x,t)|$ grew too much, it would have to be in the form of thin fingerlike structures as shown in Fig. 1 below, which tend to be very effectively dissipated by viscosity, as measured

[1] Throughout the text, vector quantities will be denoted by boldface Latin letters.

[2] More precisely, it will be sufficient to assume that f_i, g_i are continuous functions of their arguments, with continuous derivatives $\partial f_i/\partial x$, $\partial f_i/\partial u_j$, $\partial g_i/\partial u_i$, $1\le i,j\le m$.

[3] In contrast with (1.2), no growth condition is needed for $\boldsymbol{g}(t,\cdot)$, see e.g. [22] and Section 2 below.

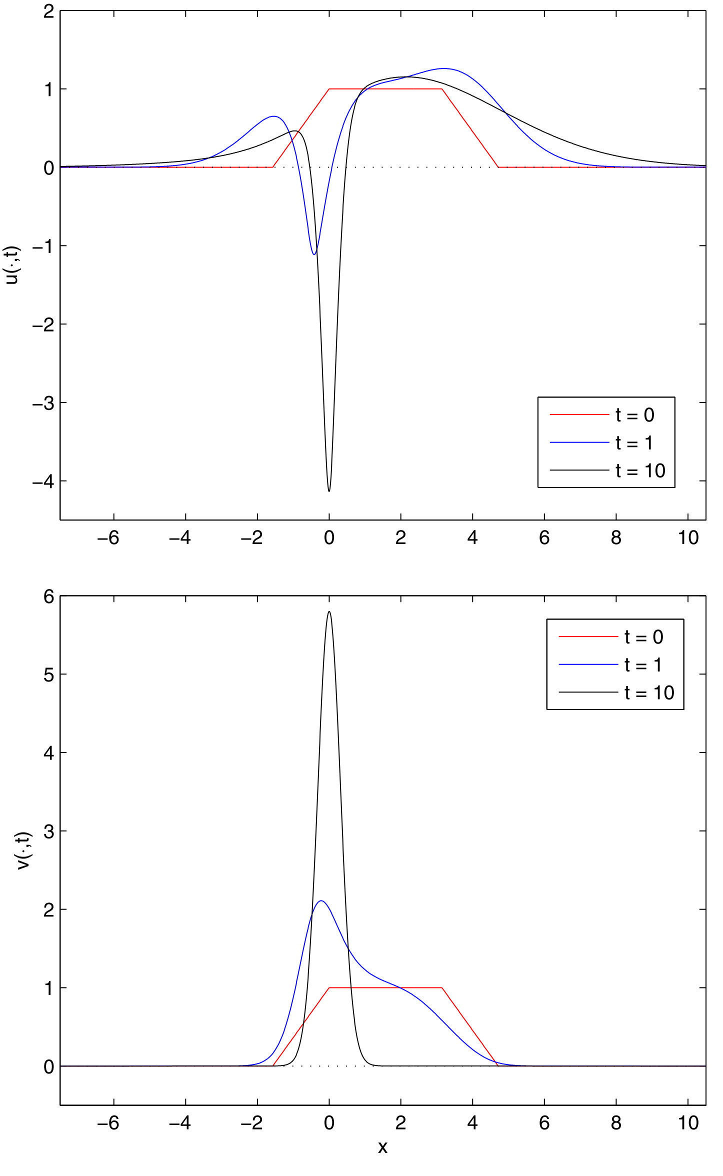

Fig. 1. Time snapshot profiles at $t = 0, 1, 10$ showing the typical fingerlike growth experienced by solutions of (1.1). Here, we consider the 2×2 system $u_t + (a(10x)v)_x = 0.4u_{xx}$, $v_t - (a(x)v)_x = 0.1v_{xx}$, where $a(x) = \text{tgh}\, x$, with $u(\cdot, t)$ displayed in the top picture, and $v(\cdot, t)$ at the bottom. The solution $v(\cdot, t)$ approaches the steady state $\mathrm{v}(x) = c\, \text{sech}^{10} x$, $c = 5.801331$ determined by mass conservation. The theoretical supnorm bound obtained by (1.7) with $p = 2$ overestimates the actual value 5.798512 by a factor 8.76 in this case. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

by $\mu(t)$. These opposing tendencies compete with each other to determine the overall solution behavior, and so they must be carefully considered before $\|u(\cdot, t)\|_{L^\infty(\mathbb{R})}$ can be estimated to any reasonable accuracy. In the case of systems with two or more equations, coupling effects may introduce further complications but the basic mechanism is still there, somehow linking solution size with lower L^p estimates and similar competing effects.

Motivated by the previous discussion, we introduce the quantities $\mu(t)$, $\mathbb{B}_\mu(0;t)$, $\mathbb{U}_p(0;t)$, $\mathcal{B}_\mu, \mathcal{U}_p$ defined by

$$\mu(t) := \min\{\mu_i(t), 1 \le i \le m\}, \quad t \ge 0, \tag{1.3}$$

$$\mathbb{B}_\mu(0;t) := \sup_{0\le\tau\le t} \frac{B(\tau)}{\mu(\tau)}, \quad t \ge 0, \tag{1.4a}$$

$$\mathcal{B}_\mu := \limsup_{t\to\infty} \frac{B(t)}{\mu(t)} \le \infty, \tag{1.4b}$$

where $B(\cdot)$ is given in (1.2) above, and

$$\mathbb{U}_p(0;t) := \sup_{0\le\tau\le t} \big\|\boldsymbol{u}(\cdot,\tau)\big\|_{L^p(\mathbb{R})}, \quad t \ge 0, \tag{1.5a}$$

$$\mathcal{U}_p := \limsup_{t\to\infty} \big\|\boldsymbol{u}(\cdot,t)\big\|_{L^p(\mathbb{R})} \le \infty. \tag{1.5b}$$

(It will only be required to assume $\mathcal{B}_\mu < \infty$ (and also that $\mathcal{U}_p < \infty$ for some $p \in [p_0,\infty[$) in the discussion in Section 3, which is aimed at Theorem B below.) To state our main results, we need for definiteness fix the meaning of the vector norms $\|\boldsymbol{u}(\cdot,t)\|_{L^\infty(\mathbb{R})}$, $\|\boldsymbol{u}(\cdot,t)\|_{L^p(\mathbb{R})}$. It will prove convenient to put

$$\big\|\boldsymbol{u}(\cdot,t)\big\|_{L^\infty(\mathbb{R})} := \max\big\{\big\|u_i(\cdot,t)\big\|_{L^\infty(\mathbb{R})}, 1 \le i \le m\big\}, \tag{1.6a}$$

where $\|u_i(\cdot,t)\|_{L^\infty(\mathbb{R})} = \sup_{x\in\mathbb{R}} |u_i(x,t)|$, and

$$\big\|\boldsymbol{u}(\cdot,t)\big\|_{L^p(\mathbb{R})} := \left\{\sum_{i=1}^m \int_{\mathbb{R}} \big|u_i(x,t)\big|^p dx\right\}^{1/p} \tag{1.6b}$$

for $1 \le p < \infty$. (Note that, adopting (1.6), we have $\|\boldsymbol{u}(\cdot,t)\|_{L^\infty(\mathbb{R})} = \lim_{p\to\infty} \|\boldsymbol{u}(\cdot,t)\|_{L^p(\mathbb{R})}$.[4]) We then have

Theorem A. *Let $\boldsymbol{u}(\cdot,t) \in L^\infty([0,T], L^\infty(\mathbb{R})^m)$ solve problem (1.1), (1.2). Then*

$$\big\|\boldsymbol{u}(\cdot,t)\big\|_{L^\infty(\mathbb{R})} \le (2p\sqrt{m})^{\frac{1}{p}} \cdot \max\big\{\big\|\boldsymbol{u}(\cdot,0)\big\|_{L^\infty(\mathbb{R})}; \mathbb{B}_\mu(0;t)^{\frac{1}{p}}\,\mathbb{U}_p(0;t)\big\} \tag{1.7}$$

for all $0 \le t \le T$, $p_0 \le p < \infty$, where $\mathbb{B}_\mu(0;t)$, $\mathbb{U}_p(0;t)$ are given in (1.4a) and (1.5a).

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[4] Moreover, it is worth noticing that, under the definitions (1.6) above, if an inequality of Gagliardo type $\|\mathsf{u}\|_{L^q} \le K\|\mathsf{u}\|_{L^{r_1}}^{1-\theta}\|\mathsf{u}_x\|_{L^{r_2}}^{\theta}$ holds for *scalar* functions u (and some appropriate constant K), then it will also be valid for *vector* functions $\boldsymbol{u}$ with the *same* constant K as in the scalar case.

It follows from Theorem A and Theorem 2.1 of Section 2 that, as previously stated, the solutions to (1.1), (1.2) are globally defined. Furthermore, we have the following general estimate for the large time size of $\boldsymbol{u}(\cdot,t)$.

Theorem B. *Let $\boldsymbol{u}(\cdot,t) \in L^{\infty}_{\mathtt{loc}}([0,\infty[, L^{\infty}(\mathbb{R})^m)$ solve problem (1.1), (1.2). Assuming that*

$$\int_0^{\infty} \mu(\tau)d\tau = \infty, \tag{1.8a}$$

then we have[5]

$$\limsup_{t\to\infty} \big\|\boldsymbol{u}(\cdot,t)\big\|_{L^\infty(\mathbb{R})} \leq \left(\frac{3\sqrt{3}}{2\pi}\sqrt{m}\,p\right)^{\frac{1}{p}} \cdot \mathcal{B}_\mu^{\frac{1}{p}} \cdot \limsup_{t\to\infty} \big\|\boldsymbol{u}(\cdot,t)\big\|_{L^p(\mathbb{R})} \tag{1.8b}$$

for every $p_0 \leq p < \infty$, where $\mu(t)$, $\mathcal{B}_\mu$ are defined in (1.3), (1.4b) above.

Theorem A is derived in Section 2, where its proof is divided into four propositions. The results obtained there are also used in Section 3, where Theorem B is established.

Example. A special, important instance of problem (1.1), (1.2) is given by the particular case $p_0 = 1$ and

$$f_i(x,t,\boldsymbol{u}) = b_i(x,t,\boldsymbol{u})u_i, \quad 1 \leq i \leq m, \tag{1.9a}$$

for $x \in \mathbb{R}$, $t \geq 0$, $\boldsymbol{u} = (u_1,\dots,u_m) \in \mathbb{R}^m$, with $b_i \in L^\infty(\mathbb{R}\times[0,\infty[\times\mathbb{R}^m)$ for each i. For these systems, it can be shown[6] that $\|u_i(\cdot,t)\|_{L^1(\mathbb{R})}$ is monotonically decreasing in $[0,\infty[$, for every i. Setting $\beta_1(t),\dots,\beta_m(t)$ given by

$$\beta_i(t) = \frac{1}{2}\Big\{\inf_{x\in\mathbb{R}} b_i\big(x,t,\boldsymbol{u}(x,t)\big) + \sup_{x\in\mathbb{R}} b_i\big(x,t,\boldsymbol{u}(x,t)\big)\Big\}, \quad 1 \leq i \leq m, \tag{1.9b}$$

and redefining $f_i(x,t,\boldsymbol{u})$ by subtracting $\beta_i(t)u_i$ from the original expression $f_i(x,t,\boldsymbol{u})$ (and correspondingly redefining $g_i(t,u_i)$ by adding $\beta_i(t)u_i$ to the original $g_i(t,u_i)$ given), we get, applying the analysis in the following sections to this particular example, the *componentwise* estimate [15]

$$\limsup_{t\to\infty} \big\|u_i(\cdot,t)\big\|_{L^\infty(\mathbb{R})} \leq \frac{3\sqrt{3}}{2\pi} \cdot \limsup_{t\to\infty} \frac{B_i(t)}{\mu_i(t)} \cdot \big\|u_i(\cdot,0)\big\|_{L^1(\mathbb{R})} \tag{1.9c}$$

if $\int_0^\infty \mu_i(\tau)d\tau = \infty$, where $B_i(t) = \frac{1}{2}\{\sup_{x\in\mathbb{R}} b_i(x,t,\boldsymbol{u}(x,t)) - \inf_{x\in\mathbb{R}} b_i(x,t,\boldsymbol{u}(x,t))\}$.

5 In (1.8b) and other similar expressions in the text, it is always understood that $0\cdot\infty = \infty\cdot 0 = \infty$. Thus, Theorem B only says something useful when $\mathcal{B}_\mu < \infty$ and $p \in [p_0,\infty[$ is such that $\mathcal{U}_p < \infty$.

6 This can be easily proved using the argument given in the proof of Theorem 2.1 below.

2. Basic estimates

This section proves Theorem A along with some auxiliary results that will be also needed in the derivation of Theorem B, which is carried out later in Section 3. (Recall that a solution on some time interval $[0, T_*[$, where $0 < T_* \leq \infty$, is a function $\boldsymbol{u}(\cdot, t) \in L^\infty_{\mathtt{loc}}([0, T_*[, L^\infty(\mathbb{R})^m)$ which is smooth (C^2 in x, C^1 in t) in $\mathbb{R} \times]0, T_*[$ and solves Eq. (1.1a) there, verifying the initial condition in the sense of $L^1_{\mathtt{loc}}(\mathbb{R})^m$, i.e., $\|\boldsymbol{u}(\cdot, t) - \boldsymbol{u}(\cdot, 0)\|_{L^1(\mathbb{K})} \to 0$ as $t \to 0$ on compact sets $\mathbb{K} \subset \mathbb{R}$. For local existence theory, see e.g. [7, Ch. 5], or [20, Ch. 6], for a short discussion that is sufficient for our needs, or [14] for a fuller treatment. By the end of this section it will be known that $T_* = \infty$.) We begin with an important elementary estimate.[7]

Theorem 2.1. *If $\boldsymbol{u}(\cdot, t) \in L^\infty_{\mathtt{loc}}([0, T_*[, L^\infty(\mathbb{R})^m)$ solves problem (1.1), (1.2) above, then $\boldsymbol{u}(\cdot, t) \in C^0([0, T_*[, L^q(\mathbb{R})^m)$ for each $q \geq \max\{p_0, 2\}$ finite, and*

$$\big\|\boldsymbol{u}(\cdot, t)\big\|_{L^q(\mathbb{R})} \leq \big\|\boldsymbol{u}(\cdot, 0)\big\|_{L^q(\mathbb{R})} \cdot \exp\left\{\frac{m}{2}(q-1)\int_0^t \frac{B(\tau)^2}{\mu(\tau)}d\tau\right\} \tag{2.1}$$

for all $0 < t < T_*$.

Proof. As the proof is relatively standard (see e.g. [5,7,13,21]), we will mostly sketch it. Taking $S \in C^1(\mathbb{R})$ such that $S'(\mathrm{v}) \geq 0$ for all v, $S(0) = 0$, $S(\mathrm{v}) = \operatorname{sgn}(\mathrm{v})$ for $|\mathrm{v}| \geq 1$, let (given $\delta > 0$) $L_\delta(\mathrm{u}) := \int_0^{\mathrm{u}} S(\mathrm{v}/\delta)d\mathrm{v}$, so that $L_\delta(\mathrm{u}) \to |\mathrm{u}|$ as $\delta \to 0$, uniformly in $\mathrm{u} \in \mathbb{R}$. Now, let $\Phi_\delta(\mathrm{u}) := L_\delta(\mathrm{u})^q$. Given $R > 0$, $0 < \epsilon \leq 1$, let $\zeta_R(\cdot)$ be the cut-off function

$$\zeta_R(x) = \begin{cases} 0, & \text{if } |x| \geq R, \\ \exp\{-\epsilon\sqrt{1+x^2}\} - \exp\{-\epsilon\sqrt{1+R^2}\}, & \text{if } |x| < R. \end{cases}$$

Fixing (any) $T \in]0, T_*[$, so that $\boldsymbol{u} \in L^\infty(\mathbb{R} \times [0, T])$, we proceed as follows. Multiplying Eq. (1.1a) by $\Phi'_\delta(u_i(x,t)) \cdot \zeta_R(x)$ if $q > 2$, or $u_i(x,t) \cdot \zeta_R(x)$ if $q = 2$, and integrating the result on $\mathbb{R} \times [0, t]$, we get (integrating by parts, summing for $1 \leq i \leq m$, and letting $\delta \to 0$):

$$\begin{aligned} &\sum_{i=1}^m \int_{|x|<R} \big|u_i(x,t)\big|^q \zeta_R(x)dx + q(q-1)\sum_{i=1}^m \int_0^t \mu_i(\tau) \int_{|x|<R} |u_i|^{q-2}\left|\frac{\partial u_i}{\partial x}\right|^2 \zeta_R(x)dx\,d\tau \\ &\quad \leq \sum_{i=1}^m \int_{|x|<R} \big|u_i(x,0)\big|^q \zeta_R(x)dx \\ &\qquad + q(q-1)\sum_{i=1}^m \int_0^t \int_{|x|<R} |u_i|^{q-2}\big|f_i(x,\tau,\boldsymbol{u})\big|\left|\frac{\partial u_i}{\partial x}\right|\zeta_R(x)dx\,d\tau \end{aligned}$$

[7] The corresponding estimate for $q = \infty$ is harder to obtain and will be given in Theorem 2.4 below.

$$+ q \sum_{i=1}^{m} \int_0^t \int_{|x|<R} |u_i|^{q-1} \big| f_i(x,\tau,\boldsymbol{u}) \big| \big| \zeta_R'(x) \big| dx d\tau$$

$$+ \sum_{i=1}^{m} \int_0^t \mu_i(\tau) \int_{|x|<R} |u_i|^{q} \big| \zeta_R''(x) \big| dx d\tau$$

$$+ \sum_{i=1}^{m} \int_0^t \mu_i(\tau) K_i(R,\tau) d\tau \cdot e^{-\epsilon R} + \sum_{i=1}^{m} \int_0^t \int_{|x|<R} \big| \mathcal{G}_i\big(\tau, u_i(x,\tau)\big) \big| \big| \zeta_R'(x) \big| dx d\tau$$

for all $0 \le t \le T$, using that $q \ge 2$ gives $L_\delta(u_i)^{q-1} L_\delta''(u_i) \to 0$ as $\delta \to 0$, with $L_\delta(u_i)^{q-1} L_\delta''(u_i)$ bounded uniformly in $\delta > 0$, and where we have also used that $\boldsymbol{u} \in L^\infty(\mathbb{R} \times [0,T])$; here, $K_i(R,\tau) = |u_i(R,\tau)|^q + |u_i(-R,\tau)|^q$, $1 \le i \le m$, and $\mathcal{G}_i(\tau,\mathrm{u})$ is given by

$$\mathcal{G}_i(\tau,\mathrm{u}) := q \int_0^{\mathrm{u}} |\mathrm{v}|^{q-1} \left| \frac{\partial g_i}{\partial u_i}(\tau,\mathrm{v}) \right| d\mathrm{v}, \quad 0 \le \tau \le T,\ \mathrm{u} \in \mathbb{R}$$

for $1 \le i \le m$. (Notice that $|\mathcal{G}_i(\tau,\mathrm{u})| \le M(T)|\mathrm{u}|^q$, for some constant $M(T)$ depending on T.) This gives, using Cauchy–Young's inequality [10, p. 622] on the third term on the right hand side and letting $R \to \infty$,

$$\sum_{i=1}^{m} \int_{\mathbb{R}} \big| u_i(x,t) \big|^q w_\epsilon(x) dx + \frac{1}{2} q(q-1) \sum_{i=1}^{m} \int_0^t \mu_i(\tau) \int_{\mathbb{R}} \big| u_i(x,\tau) \big|^{q-2} \left| \frac{\partial u_i}{\partial x} \right|^2 w_\epsilon(x) dx d\tau$$

$$\le \sum_{i=1}^{m} \int_{\mathbb{R}} \big| u_i(x,0) \big|^q w_\epsilon(x) dx + \frac{1}{2} q(q-1) \sum_{i=1}^{m} \int_0^t \mu(\tau)^{-1} \int_{\mathbb{R}} |u_i|^{q-2} \big| f_i(x,\tau,\boldsymbol{u}) \big|^2 w_\epsilon(x) dx d\tau$$

$$+ q\epsilon \sum_{i=1}^{m} \int_0^t \int_{\mathbb{R}} |u_i|^{q-1} \big| f_i(x,\tau,\boldsymbol{u}) \big| w_\epsilon(x) dx d\tau$$

$$+ \epsilon \sum_{i=1}^{m} \int_0^t \big[\mu_i(\tau) + M(T) \big] \int_{\mathbb{R}} \big| u_i(x,\tau) \big|^q w_\epsilon(x) dx d\tau$$

for all $0 \le t \le T$, with $\mu(t)$ given in (1.3), since we have $|\zeta_R'(x)| \le \epsilon w_\epsilon(x)$, $|\zeta_R''(x)| \le \epsilon w_\epsilon(x)$, where $w_\epsilon(x) := \exp\{-\epsilon\sqrt{1+x^2}\}$. Observing that, by (1.2) and Hölder's inequality, we have

$$\sum_{i=1}^{m} \big| u_i(x,t) \big|^{q-2} \big| f_i(x,t,\boldsymbol{u}) \big|^2 \le \big| \boldsymbol{f}(x,t,\boldsymbol{u}) \big|_2^2 \sum_{i=1}^{m} \big| u_i(x,t) \big|^{q-2} \le m B(t)^2 \sum_{i=1}^{m} \big| u_i(x,t) \big|^q,$$

$$\sum_{i=1}^{m} \big| u_i(x,t) \big|^{q-1} \big| f_i(x,t,\boldsymbol{u}) \big| \le \big| \boldsymbol{f}(x,t,\boldsymbol{u}) \big|_2 \sum_{i=1}^{m} \big| u_i(x,t) \big|^{q-1} \le \sqrt{m} B(t) \sum_{i=1}^{m} \big| u_i(x,t) \big|^q,$$

where $|\cdot|_2$ denotes the Euclidean norm, we then obtain

$$\mathrm{U}_\epsilon(t) + V_\epsilon(t) \le \mathrm{U}_\epsilon(0) + \int_0^t G_\epsilon(\tau)\mathrm{U}_\epsilon(\tau)d\tau, \qquad \mathrm{U}_\epsilon(t) = \sum_{i=1}^m \int_{\mathbb{R}} \left|u_i(x,t)\right|^q w_\epsilon(x)dx, \tag{2.2a}$$

where $G_\epsilon(t) = \frac{m}{2}q(q-1)\frac{B(t)^2}{\mu(t)} + \epsilon mqB(t) + \epsilon M(T) + \epsilon\sum_{i=1}^m \mu_i(t)$, and

$$\mathrm{V}_\epsilon(t) = \frac{1}{2}q(q-1)\sum_{i=1}^m \int_0^t \mu_i(\tau)\int_{\mathbb{R}} \left|u_i(x,\tau)\right|^{q-2}\left|\frac{\partial u_i}{\partial x}\right|^2 w_\epsilon(x)dxd\tau. \tag{2.2b}$$

By Gronwall's lemma, (2.2) gives $\mathrm{U}_\epsilon(t) \le \mathrm{U}_\epsilon(0)\cdot\exp\{\int_0^t G_\epsilon(\tau)d\tau\}$, from which we obtain (2.1) by simply letting $\epsilon \to 0$. This shows, in particular, that $\boldsymbol{u}(\cdot,t) \in L^\infty_{\mathrm{loc}}([0,T_*[, L^q(\mathbb{R})^m)$ if $\max\{p_0, 2\} \le q < \infty$. Now, to get that $\boldsymbol{u}(\cdot,t) \in C^0([0,T_*[, L^q(\mathbb{R})^m)$, it is sufficient to show that, given $\varepsilon > 0$ and $0 < T < T_*$ arbitrary, we can find $R = R(\varepsilon, T) \gg 1$ large enough so that we have $\|\boldsymbol{u}(\cdot,t)\|_{L^q(|x|>R)} < \varepsilon$ for any $0 \le t \le T$. Taking $\psi \in C^2(\mathbb{R})$ with $0 \le \psi \le 1$ and $\psi(x) = 0$ for all $x \le 0$, $\psi(x) = 1$ for all $x \ge 1$, let $\Psi_{R,M} \in C^2(\mathbb{R})$ be the cut-off function given by $\Psi_{R,M}(x) = 0$ if $|x| \le R-1$, $\Psi_{R,M}(x) = \psi(|x| - R + 1)$ if $R - 1 < |x| < R$, and $\Psi_{R,M}(x) = 1$ if $R \le |x| \le R + M$, $\Psi_{R,M}(x) = \psi(R + M + 1 - |x|)$ if $R + M < |x| < R + M + 1$, $\Psi_{R,M}(x) = 0$ if $|x| \ge R + M + 1$, where $R > 1$, $M > 0$ are given (and arbitrary). Multiplying Eq. (1.1a) by $\Phi'_\delta(u_i(x,t))\cdot\Psi_{R,M}(x)$ if $q > 2$, or $u_i(x,t)\cdot\Psi_{R,M}(x)$ if $q = 2$, integrating the result on $\mathbb{R}\times[0,t]$, $0 < t \le T$, and summing for $1 \le i \le m$, we obtain, as in (2.2), by letting $\delta \to 0$, $M \to \infty$, that $\|\boldsymbol{u}(\cdot,t)\|_{L^q(|x|>R)} < \varepsilon/2 + \|\boldsymbol{u}(\cdot,0)\|_{L^q(|x|>R-1)}$ for all $0 \le t \le T$, provided that we take $R > 1$ sufficiently large. This gives the continuity result, and the proof is complete. □

Remark. It follows from the proof of Theorem 2.1 above that both properties (2.1) and $\boldsymbol{u}(\cdot,t) \in C^0([0,T_*[, L^q(\mathbb{R})^m)$ hold for any $p_0 \le q < \infty$ when $\boldsymbol{f}$ has the form (1.9a).

An important by-product of the proof above is that we have (letting $\epsilon \to 0$ in (2.2), and using (2.1)), for each $0 < T < T_*$ and $\max\{p_0, 2\} \le q < \infty$,

$$\sum_{i=1}^m \int_0^T \int_{\mathbb{R}} \left|u_i(x,\tau)\right|^{q-2}\left|\frac{\partial u_i}{\partial x}\right|^2 dxd\tau < \infty. \tag{2.3}$$

Therefore, if we repeat the steps above leading to (2.2), but without majorizing, we obtain (letting $\delta \to 0$, $R \to \infty$, $\epsilon \to 0$, in this order, taking (2.1), (2.3) into account) the energy *identity*

$$\begin{aligned}
&\|\boldsymbol{u}(\cdot,t)\|^q_{L^q(\mathbb{R})} + q(q-1)\sum_{i=1}^m \int_0^t \mu_i(\tau)\int_{\mathbb{R}} \left|u_i(x,\tau)\right|^{q-2}\left|\frac{\partial u_i}{\partial x}\right|^2 dxd\tau \\
&\quad = \|\boldsymbol{u}(\cdot,0)\|^q_{L^q(\mathbb{R})} + q(q-1)\sum_{i=1}^m \int_0^t \int_{\mathbb{R}} \left|u_i(x,\tau)\right|^{q-2} f_i(x,\tau,\boldsymbol{u})\frac{\partial u_i}{\partial x}dxd\tau
\end{aligned} \tag{2.4}$$

for all $0 < t < T_*$. The core of the difficulty in the analysis of the problem (1.1a), (1.1b) is already apparent here: under the general condition (1.2), it is not very clear how one should go about the last term in (2.4) in order to get more than (2.1). Actually, it will be convenient to consider (2.4) in the (equivalent) differential form, i.e.,

$$\frac{d}{dt}\|\boldsymbol{u}(\cdot,t)\|^q_{L^q(\mathbb{R})} + q(q-1)\sum_{i=1}^m \mu_i(t)\int_{\mathbb{R}} |u_i(x,t)|^{q-2}\left|\frac{\partial u_i}{\partial x}\right|^2 dx$$
$$= q(q-1)\sum_{i=1}^m \int_{\mathbb{R}} |u_i(x,t)|^{q-2} f_i(x,t,\boldsymbol{u})\frac{\partial u_i}{\partial x}dx, \tag{2.5}$$

for all $t \in [0, T_*[\setminus E_q$, where $E_q \subset [0, T_*[$ has zero measure. (This follows from (2.4) by applying Lebesgue's differentiation theorem.) Recalling the Nash inequality [16] in 1-D,

$$\|\mathrm{v}\|_{L^2(\mathbb{R})} \le C_2\|\mathrm{v}\|^{2/3}_{L^1(\mathbb{R})}\|\mathrm{v}_x\|^{1/3}_{L^2(\mathbb{R})}, \quad C_2 = \left(\frac{3\sqrt{3}}{4\pi}\right)^{1/3}, \tag{2.6}$$

where the value given above for C_2 is optimal [6], and the one-dimensional inequality

$$\|\mathrm{v}\|_{L^\infty(\mathbb{R})} \le C_\infty\|\mathrm{v}\|^{1/3}_{L^1(\mathbb{R})}\|\mathrm{v}_x\|^{2/3}_{L^2(\mathbb{R})}, \quad C_\infty = \left(\frac{3}{4}\right)^{2/3}, \tag{2.7}$$

valid for any $\mathrm{v} \in L^1(\mathbb{R}) \cap H^1(\mathbb{R})$, we get the following basic consequence of (2.5)–(2.7).

Theorem 2.2. *Let $2p_0 \le q < \infty$. If $\hat{t} \in [0, T_*[\setminus E_q$ is such that $\frac{d}{dt}\|\boldsymbol{u}(\cdot,t)\|^q_{L^q(\mathbb{R})}|_{t=\hat{t}} \ge 0$, then*

$$\|\boldsymbol{u}(\cdot,\hat{t})\|_{L^q(\mathbb{R})} \le \left(\frac{q}{2}\sqrt{m}C_2^3\right)^{1/q}\left(\frac{B(\hat{t})}{\mu(\hat{t})}\right)^{1/q}\|\boldsymbol{u}(\cdot,\hat{t})\|_{L^{q/2}(\mathbb{R})} \tag{2.8a}$$

and

$$\|\boldsymbol{u}(\cdot,\hat{t})\|_{L^\infty(\mathbb{R})} \le \left(\frac{q}{2}\sqrt{m}C_2C_\infty\right)^{2/q}\left(\frac{B(\hat{t})}{\mu(\hat{t})}\right)^{2/q}\|\boldsymbol{u}(\cdot,\hat{t})\|_{L^{q/2}(\mathbb{R})}, \tag{2.8b}$$

where C_2, C_∞ are the constants given in (2.6), (2.7), and B, μ are defined in (1.2), (1.3).

Proof. Consider the bound (2.8a) first. Using the hypothesis and (2.5), we get, for $q \ge 2p_0$, with the help of Cauchy–Schwarz's inequality,

$$\sum_{i=1}^m \mu_i(\hat{t})\int_{\mathbb{R}} |u_i(x,\hat{t})|^{q-2}\left|\frac{\partial u_i}{\partial x}\right|^2 dx \le \frac{1}{\mu(\hat{t})}\sum_{i=1}^m\int_{\mathbb{R}} |u_i(x,\hat{t})|^{q-2}|f_i(x,\hat{t},\boldsymbol{u})|^2 dx$$
$$\le \frac{1}{\mu(\hat{t})} m B(\hat{t})^2 \|\boldsymbol{u}(\cdot,\hat{t})\|^q_{L^q(\mathbb{R})},$$

recalling (1.2), (1.3). Hence, we have

$$\sum_{i=1}^{m} \int_{\mathbb{R}} |u_i(x,\hat{t})|^{q-2} \left|\frac{\partial u_i}{\partial x}\right|^2 dx \leq m \cdot \left(\frac{B(\hat{t})}{\mu(\hat{t})}\right)^2 \|\boldsymbol{u}(\cdot,\hat{t})\|_{L^q(\mathbb{R})}^q.$$

In terms of the function $\boldsymbol{v} = (v_1, \ldots, v_m) \in L^1(\mathbb{R}) \cap L^\infty(\mathbb{R})$ defined by $v_i(x) := |u_i(x,\hat{t})|^{q/2}$ if $q > 2$, and $v_i(x) := u_i(x,\hat{t})$ if $q = 2$, for each $1 \leq i \leq m$, the inequality above reads

$$\|\boldsymbol{v}_x\|_{L^2(\mathbb{R})} \leq \frac{q}{2}\sqrt{m}\frac{B(\hat{t})}{\mu(\hat{t})}\|\boldsymbol{v}\|_{L^2(\mathbb{R})}.$$

Using (2.6), we then get $\|\boldsymbol{v}\|_{L^2(\mathbb{R})}^2 \leq \frac{q}{2}\sqrt{m}C_2^3\frac{B(\hat{t})}{\mu(\hat{t})}\|\boldsymbol{v}\|_{L^1(\mathbb{R})}^2$, which is equivalent to (2.8a). Similarly, (2.8b) is obtained, using (2.7). $\square$

Thus, we can use (2.8) when $\|\boldsymbol{u}(\cdot,t)\|_{L^q(\mathbb{R})}$ is not decreasing. If it is decreasing, (2.5) becomes useless but at least we know in such case that $\|\boldsymbol{u}(\cdot,t)\|_{L^q(\mathbb{R})}$ is not increasing, which should be useful too. Different values of q have different scenarios, which have to be pieced together in some way. The next result shows us just how. To this end, it is convenient to introduce (as in (1.4), (1.5)) the quantities $\mathbb{B}_\mu(t_0;t)$, $\mathbb{U}_p(t_0;t)$ defined by

$$\mathbb{B}_\mu(t_0;t) = \sup\left\{\frac{B(\tau)}{\mu(\tau)} : t_0 \leq \tau \leq t\right\}, \tag{2.9}$$

$$\mathbb{U}_p(t_0;t) = \sup\left\{\left\|\boldsymbol{u}(\cdot,\tau)\right\|_{L^p(\mathbb{R})} : t_0 \leq \tau \leq t\right\}, \tag{2.10}$$

given $p \geq p_0$, $0 \leq t_0 \leq t < T_*$ arbitrary.

Theorem 2.3. *Let $q \geq 2p_0$. For each $0 \leq t_0 < T_*$, we have*

$$\mathbb{U}_q(t_0;t) \leq \max\left\{\left\|\boldsymbol{u}(\cdot,t_0)\right\|_{L^q(\mathbb{R})};\ \left(\frac{q}{2}\sqrt{m}C_2^3\right)^{\frac{1}{q}}\mathbb{B}_\mu(t_0;t)^{\frac{1}{q}}\,\mathbb{U}_{\frac{q}{2}}(t_0;t)\right\} \tag{2.11}$$

for all $t_0 \leq t < T_*$.

Proof. Set $\lambda_q(t) = (\frac{q}{2}\sqrt{m}C_2^3)^{\frac{1}{q}}\mathbb{B}_\mu(t_0;t)^{\frac{1}{q}}\mathbb{U}_{\frac{q}{2}}(t_0;t)$. There are three cases to consider.

`Case I:` $\|\boldsymbol{u}(\cdot,\tau)\|_{L^q(\mathbb{R})} > \lambda_q(t)$ for all $t_0 \leq \tau \leq t$. By (2.8a), Theorem 2.2, we must then have $d/d\tau\|\boldsymbol{u}(\cdot,\tau)\|_{L^q(\mathbb{R})}^q < 0$ for all $\tau \in [t_0,t] \setminus E_q$, so that $\|\boldsymbol{u}(\cdot,\tau)\|_{L^q(\mathbb{R})}$ is monotonically decreasing in $[t_0,t]$. In particular, $\mathbb{U}_q(t_0;t) = \|\boldsymbol{u}(\cdot,t_0)\|_{L^q(\mathbb{R})}$ in this case.

`Case II:` $\|\boldsymbol{u}(\cdot,t_0)\|_{L^q(\mathbb{R})} > \lambda_q(t)$ and $\|\boldsymbol{u}(\cdot,t_1)\|_{L^q(\mathbb{R})} \leq \lambda_q(t)$ for some $t_1 \in\,]t_0,t]$.

In this case, let $t_2 \in\,]t_0,t]$ be such that we have $\|\boldsymbol{u}(\cdot,\tau)\|_{L^q(\mathbb{R})} > \lambda_q(t)$ for all $t_0 \leq \tau < t_2$, while $\|\boldsymbol{u}(\cdot,t_2)\|_{L^q(\mathbb{R})} = \lambda_q(t)$. We claim that $\|\boldsymbol{u}(\cdot,\tau)\|_{L^q(\mathbb{R})} \leq \lambda_q(t)$ for every $t_2 \leq \tau \leq t$: in fact, if this were not true, we could then find t_3, t_4 with $t_2 \leq t_3 < t_4 \leq t$ such that $\|\boldsymbol{u}(\cdot,\tau)\|_{L^q(\mathbb{R})} > \lambda_q(t)$ for all $t_3 < \tau \leq t_4$, $\|\boldsymbol{u}(\cdot,t_3)\|_{L^q(\mathbb{R})} = \lambda_q(t)$. By (2.8a), Theorem 2.2, this would require $d/d\tau\|\boldsymbol{u}(\cdot,\tau)\|_{L^q(\mathbb{R})}^q < 0$ for all $\tau \in\,]t_3,t_4] \setminus E_q$, so that $\|\boldsymbol{u}(\cdot,\tau)\|_{L^q(\mathbb{R})}$ could not increase anywhere on $[t_3, t_4]$. This contradicts $\|\boldsymbol{u}(\cdot, t_3)\|_{L^q(\mathbb{R})} < \|\boldsymbol{u}(\cdot, t_4)\|_{L^q(\mathbb{R})}$, and so we have $\|\boldsymbol{u}(\cdot, \tau)\|_{L^q(\mathbb{R})} \le \lambda_q(t)$ for every $t_2 \le \tau \le t$, as claimed. On the other hand, by (2.8a), $\|\boldsymbol{u}(\cdot, \tau)\|_{L^q(\mathbb{R})}$ has to be monotonically decreasing on $[t_0, t_2]$, just as in `Case I`. Therefore, we have $\mathbb{U}_q(t_0; t) = \|\boldsymbol{u}(\cdot, t_0)\|_{L^q(\mathbb{R})}$ in this case again.

`Case III:` $\|\boldsymbol{u}(\cdot, t_0)\|_{L^q(\mathbb{R})} \le \lambda_q(t)$. This gives $\|\boldsymbol{u}(\cdot, \tau)\|_{L^q(\mathbb{R})} \le \lambda_q(t)$ for every $t_0 \le \tau \le t$, by repeating the argument used on the interval $[t_2, t]$ in `Case II` above. It follows that we have $\mathbb{U}_q(t_0; t) \le \lambda_q(t)$ in this case, and the proof of Theorem 2.3 is complete. □

An important consequence of Theorem 2.3 is that we can finally show Theorem A, which for convenience is restated as Theorem 2.4 below.

Theorem 2.4. *Let $p_0 \le p < \infty$, $0 \le t_0 < T_*$. Then*

$$\big\|\boldsymbol{u}(\cdot, t)\big\|_{L^\infty(\mathbb{R})} \le (2p\sqrt{m})^{\frac{1}{p}} \cdot \max\big\{\big\|\boldsymbol{u}(\cdot, t_0)\big\|_{L^\infty(\mathbb{R})}; \mathbb{B}_\mu(t_0; t)^{\frac{1}{p}}\,\mathbb{U}_p(t_0; t)\big\} \tag{2.12}$$

for any $t_0 \le t < T_*$, where $\mathbb{B}_\mu(t_0; t)$, $\mathbb{U}_p(t_0; t)$ are given in (2.9), (2.10) above.

Proof. Let $k \in \mathbb{Z}$, $k \ge 2$. Applying (2.11) successively with $q = 2p, 4p, \dots, 2^k p$, we obtain

$$\begin{aligned}\big\|\boldsymbol{u}(\cdot, t)\big\|_{L^{2^k p}(\mathbb{R})} \le \max\big\{&\big\|\boldsymbol{u}(\cdot, t_0)\big\|_{L^{2^k p}(\mathbb{R})}; K(k, \ell)^{\frac{1}{p}} \cdot \mathbb{B}_\mu(t_0; t)^{\frac{1}{p}(2^{-\ell} - 2^{-k})} \cdot \big\|\boldsymbol{u}(\cdot, t_0)\big\|_{L^{2^\ell p}(\mathbb{R})},\\ &1 \le \ell \le k-1;\ K(k, 0)^{\frac{1}{p}} \cdot \mathbb{B}_\mu(t_0; t)^{\frac{1}{p}(1 - 2^{-k})} \cdot \mathbb{U}_p(t_0; t)\big\},\end{aligned} \tag{2.13a}$$

where

$$K(k, \ell) = \prod_{j=\ell+1}^{k} \big(2^{j-1} p\sqrt{m}\, C_2^3\big)^{2^{-j}}, \quad 0 \le \ell \le k-1. \tag{2.13b}$$

Now, for $1 \le \ell \le k-1$:

$$\begin{aligned}&\mathbb{B}_\mu(t_0; t)^{\frac{1}{p}(2^{-\ell} - 2^{-k})} \cdot \big\|\boldsymbol{u}(\cdot, t_0)\big\|_{L^{2^\ell p}(\mathbb{R})}\\ &\le \mathbb{B}_\mu(t_0; t)^{\frac{1}{p}(2^{-\ell} - 2^{-k})} \cdot \big\|\boldsymbol{u}(\cdot, t_0)\big\|_{L^p(\mathbb{R})}^{\frac{2^{-\ell} - 2^{-k}}{1 - 2^{-k}}} \cdot \big\|\boldsymbol{u}(\cdot, t_0)\big\|_{L^{2^k p}(\mathbb{R})}^{\frac{1 - 2^{-\ell}}{1 - 2^{-k}}}\\ &\le \max\big\{\big\|\boldsymbol{u}(\cdot, t_0)\big\|_{L^{2^k p}(\mathbb{R})}; \mathbb{B}_\mu(t_0; t)^{\frac{1}{p}(1 - 2^{-k})} \cdot \big\|\boldsymbol{u}(\cdot, t_0)\big\|_{L^p(\mathbb{R})}\big\}\end{aligned}$$

by Young's inequality (see e.g. [10, p. 622]), so that we get, from (2.13),

$$(*) \quad \big\|\boldsymbol{u}(\cdot, t)\big\|_{L^{2^k p}(\mathbb{R})} \le (2p\sqrt{m})^{\frac{1}{p}} \cdot \max\big\{\big\|\boldsymbol{u}(\cdot, t_0)\big\|_{L^{2^k p}(\mathbb{R})}; \mathbb{B}_\mu(t_0; t)^{\frac{1}{p}(1 - 2^{-k})} \cdot \mathbb{U}_p(t_0; t)\big\},$$

since $K(k, \ell) \le 2p\sqrt{m}$ for all $0 \le \ell \le k-1$. Letting $k \to \infty$, (2.12) is obtained. □

It follows from Theorems 2.1 and 2.4 that local solutions can be continually extended so as to become globally defined (i.e., $T_* = \infty$). Going back to (∗) above, we then obtain (letting $k \to \infty$ and then $t \to \infty$):

$$\limsup_{t\to\infty} \big\| \boldsymbol{u}(\cdot,t) \big\|_{L^\infty(\mathbb{R})} \le (2p\sqrt{m})^{\frac{1}{p}} \cdot \max\big\{ \big\| \boldsymbol{u}(\cdot,t_0) \big\|_{L^\infty(\mathbb{R})}; \mathbb{B}_\mu(t_0)^{\frac{1}{p}} \mathbb{U}_p(t_0) \big\} \tag{2.14}$$

for any $t_0 \ge 0$, where $\mathbb{B}_\mu(t_0)$, $\mathbb{U}_p(t_0)$ are given by

$$\mathbb{B}_\mu(t_0) = \sup\left\{ \frac{B(t)}{\mu(t)} : t \ge t_0 \right\}, \tag{2.15}$$

$$\mathbb{U}_p(t_0) = \sup\big\{ \big\| \boldsymbol{u}(\cdot,t) \big\|_{L^p(\mathbb{R})} : t \ge t_0 \big\}. \tag{2.16}$$

Choosing $t_0 = t_0^{(n)}$ such that $t_0^{(n)} \to \infty$ and $\|\boldsymbol{u}(\cdot, t_0^{(n)})\|_{L^\infty(\mathbb{R})} \to \liminf_{t\to\infty} \|\boldsymbol{u}(\cdot,t)\|_{L^\infty(\mathbb{R})}$, and applying (2.14) for each n, we obtain, letting $n \to \infty$,

$$\limsup_{t\to\infty} \big\| \boldsymbol{u}(\cdot,t) \big\|_{L^\infty(\mathbb{R})} \le (2p\sqrt{m})^{\frac{1}{p}} \cdot \max\Big\{ \liminf_{t\to\infty} \big\| \boldsymbol{u}(\cdot,t) \big\|_{L^\infty(\mathbb{R})}; \mathcal{B}_\mu^{\frac{1}{p}} \cdot \mathcal{U}_p \Big\}, \tag{2.17}$$

where $\mathcal{B}_\mu = \limsup_{t\to\infty} \frac{B(t)}{\mu(t)}$, $\mathcal{U}_p = \limsup_{t\to\infty} \|\boldsymbol{u}(\cdot,t)\|_{L^p(\mathbb{R})}$, see (1.4b), (1.5b) in Section 1.

3. Large time estimates

In this section, we derive a few large time estimates for solutions $\boldsymbol{u}(\cdot,t)$ of (1.1a), (1.1b), which will ultimately lead to Theorem B, restated here as Theorem 3.3 below. The derivation of time asymptotic estimates for $\boldsymbol{u}(\cdot,t)$ requires, in addition to the basic condition (1.2), that

$$\int_0^\infty \mu(\tau)\,d\tau = \infty, \tag{3.1}$$

where $\mu \in C^0([0,\infty[)$ is defined in (1.3). Recalling the quantities $\mathcal{B}_\mu, \mathcal{U}_p \ge 0$ given by

$$\mathcal{B}_\mu = \limsup_{t\to\infty} \frac{B(t)}{\mu(t)}, \qquad \mathcal{U}_p = \limsup_{t\to\infty} \big\| \boldsymbol{u}(\cdot,t) \big\|_{L^p(\mathbb{R})}, \tag{3.2}$$

it is assumed throughout this section that $\mathcal{B}_\mu < \infty$. Our starting point is (2.17) above.

Theorem 3.1. *Let $\mathcal{B}_\mu < \infty$. If (3.1) holds, then*

$$\liminf_{t\to\infty} \big\| \boldsymbol{u}(\cdot,t) \big\|_{L^\infty(\mathbb{R})} \le (p\sqrt{m}C_2C_\infty)^{\frac{1}{p}} \cdot \mathcal{B}_\mu^{\frac{1}{p}} \cdot \limsup_{t\to\infty} \big\| \boldsymbol{u}(\cdot,t) \big\|_{L^p(\mathbb{R})} \tag{3.3}$$

for all $p_0 \le p < \infty$ with $\mathcal{U}_p < \infty$, where C_2, C_∞ are the constants given in (2.6), (2.7).

Proof. Given $p \in [p_0, \infty[$ such that $\mathcal{U}_p < \infty$, let $\boldsymbol{v} = (v_1, \dots, v_m) \in L^\infty(\mathbb{R} \times [0, \infty[)$ be defined by $v_i(x,t) := |u_i(x,t)|^p$ if $p > 1$, $v_i(x,t) := u_i(x,t)$ if $p = 1$, for $1 \le i \le m$. It follows that

$$\left\| \boldsymbol{v}(\cdot,t) \right\|_{L^2(\mathbb{R})}^2 = \left\| \boldsymbol{u}(\cdot,t) \right\|_{L^{2p}(\mathbb{R})}^{2p},$$

$$\left\| \boldsymbol{v}_x(\cdot,t) \right\|_{L^2(\mathbb{R})}^2 = p^2 \sum_{i=1}^m \int_{\mathbb{R}} \left| u_i(x,t) \right|^{2p-2} \left| \frac{\partial u_i}{\partial x} \right|^2 dx$$

and, by (1.2),

$$\sum_{i=1}^m \int_{\mathbb{R}} \left| u_i(x,t) \right|^{2p-2} \left| f_i(x,t,\boldsymbol{u}) \right| \left| \frac{\partial u_i}{\partial x} \right| dx \le \frac{\sqrt{m}}{p} B(t) \left\| \boldsymbol{v}(\cdot,t) \right\|_{L^2(\mathbb{R})} \left\| \boldsymbol{v}_x(\cdot,t) \right\|_{L^2(\mathbb{R})}.$$

Therefore, from (1.4a), (2.5), we have, for some null set $E_{2p} \subset [0, \infty[$,

$$\begin{aligned} &\frac{d}{dt} \left\| \boldsymbol{v}(\cdot,t) \right\|_{L^2(\mathbb{R})}^2 + 4\left(1 - \frac{1}{2p}\right) \mu(t) \left\| \boldsymbol{v}_x(\cdot,t) \right\|_{L^2(\mathbb{R})}^2 \\ &\quad \le 4p\sqrt{m}\left(1 - \frac{1}{2p}\right) B(t) \left\| \boldsymbol{v}(\cdot,t) \right\|_{L^2(\mathbb{R})} \left\| \boldsymbol{v}_x(\cdot,t) \right\|_{L^2(\mathbb{R})} \end{aligned}$$

for all $t \in [0, \infty[\setminus E_{2p}$. So, by Nash's inequality (2.6), we have

$$\begin{aligned} &\frac{d}{dt} \left\| \boldsymbol{v}(\cdot,t) \right\|_{L^2(\mathbb{R})}^2 + 4\left(1 - \frac{1}{2p}\right) \mu(t) \left\| \boldsymbol{v}_x(\cdot,t) \right\|_{L^2(\mathbb{R})}^2 \\ &\quad \le 4p\sqrt{m}C_2\left(1 - \frac{1}{2p}\right) B(t) \left\| \boldsymbol{v}(\cdot,t) \right\|_{L^1(\mathbb{R})}^{2/3} \left\| \boldsymbol{v}_x(\cdot,t) \right\|_{L^2(\mathbb{R})}^{4/3}. \end{aligned}$$

This gives, by Young's inequality [10, p. 622], for all $t \in [0, \infty[\setminus E_{2p}$,

$$\begin{aligned} &\frac{d}{dt} \left\| \boldsymbol{v}(\cdot,t) \right\|_{L^2(\mathbb{R})}^2 + \frac{4}{3}\left(1 - \frac{1}{2p}\right) \mu(t) \left\| \boldsymbol{v}_x(\cdot,t) \right\|_{L^2(\mathbb{R})}^2 \\ &\quad \le \frac{4}{3}\left(1 - \frac{1}{2p}\right) (p\sqrt{m}C_2)^3 \frac{B(t)^3}{\mu(t)^2} \left\| \boldsymbol{v}(\cdot,t) \right\|_{L^1(\mathbb{R})}^2. \end{aligned} \tag{3.4}$$

Now, letting $\lambda_p \in \mathbb{R}$, $g \in L^\infty([0, \infty[)$ be given by

$$\lambda_p = \limsup_{t \to \infty} g(t), \qquad g(t) = p\sqrt{m}C_2 \frac{B(t)}{\mu(t)} \left\| \boldsymbol{v}(\cdot,t) \right\|_{L^1(\mathbb{R})},$$

we observe that (3.3) is obtained if we show that

$$\liminf_{t \to \infty} \left\| \boldsymbol{v}(\cdot,t) \right\|_{L^\infty(\mathbb{R})} \le C_\infty \cdot \lambda_p. \tag{3.5}$$

We argue by contradiction and assume that (3.5) is false. Taking then $0 < \eta \ll 1$, $t_0 \gg 1$ so that $\|\boldsymbol{v}(\cdot,t)\|_{L^\infty(\mathbb{R})} \geq C_\infty \cdot (\lambda_p + \eta)$ and $g(t) \leq \lambda_p + \eta/2$ hold for all $t \geq t_0$, we get, by (2.7) and (3.4),

$$\begin{aligned}\|\boldsymbol{v}(\cdot,t)\|^3_{L^\infty(\mathbb{R})} &\leq C^3_\infty \|\boldsymbol{v}(\cdot,t)\|_{L^1(\mathbb{R})} \|\boldsymbol{v}_x(\cdot,t)\|^2_{L^2(\mathbb{R})} \\ &\leq C^3_\infty g(t)^3 + C^3_\infty \frac{2p}{2p-1}\frac{1}{\mu(t)}\|\boldsymbol{v}(\cdot,t)\|_{L^1(\mathbb{R})}\left(-\frac{d}{dt}\|\boldsymbol{v}(\cdot,t)\|^2_{L^2(\mathbb{R})}\right)\end{aligned}$$

for all $t \in [t_0, \infty[\setminus E_{2p}$. Since $\|\boldsymbol{v}(\cdot,t)\|_{L^\infty(\mathbb{R})} \geq C_\infty \cdot (\lambda_p + \eta)$, $g(t) \leq \lambda_p + \eta/2$, this gives

$$-\frac{d}{dt}\|\boldsymbol{v}(\cdot,t)\|^2_{L^2(\mathbb{R})} \geq K(\eta)\cdot\mu(t), \quad \forall t \in [t_0, \infty[\setminus E_{2p}$$

for some constant $K(\eta) > 0$ independent of t. In particular, it follows that we have

$$\|\boldsymbol{v}(\cdot,t_0)\|^2_{L^2(\mathbb{R})} \geq K(\eta)\int_{t_0}^{t}\mu(\tau)d\tau, \quad \forall t \geq t_0, \tag{3.6}$$

which is absurd in view of our assumption (3.1). This contradiction establishes (3.5) above. As (3.3) follows readily, the proof of Theorem 3.1 is complete. □

Another result along these lines is given next.

Theorem 3.2. *Let $\mathcal{B}_\mu < \infty$. If (3.1) holds, then*

$$\limsup_{t\to\infty}\|\boldsymbol{u}(\cdot,t)\|_{L^q(\mathbb{R})} \leq \left(\frac{q}{2}\sqrt{m}C^3_2\right)^{\frac{1}{q}} \cdot \mathcal{B}_\mu^{\frac{1}{q}} \cdot \limsup_{t\to\infty}\|\boldsymbol{u}(\cdot,t)\|_{L^{q/2}(\mathbb{R})} \tag{3.7}$$

for every $q \in [2p_0, \infty[$ with $\mathcal{U}_{q/2} < \infty$, where $C_2 = (3\sqrt{3}/(4\pi))^{1/3}$ is given in (2.6).

Proof. Let $p = q/2$. As in the previous proof, we take $\boldsymbol{v} = (v_1, \dots, v_m) \in L^\infty(\mathbb{R} \times [0, \infty[)$ given by $v_i(x,t) = |u_i(x,t)|^p$ if $p > 1$, $v_i(x,t) = u_i(x,t)$ if $p = 1$, $1 \leq i \leq m$. Setting

$$\lambda_p = \limsup_{t\to\infty} g(t), \quad g(t) = \left(p\sqrt{m}C^3_2\right)^{1/2}\left(\frac{B(t)}{\mu(t)}\right)^{1/2}\|\boldsymbol{v}(\cdot,t)\|_{L^1(\mathbb{R})},$$

we claim that

$$\limsup_{t\to\infty}\|\boldsymbol{v}(\cdot,t)\|_{L^2(\mathbb{R})} \leq \lambda_p. \tag{3.8}$$

Again, we argue by contradiction. If (3.8) is false, we can pick $0 < \eta \ll 1$ and a sequence $(t_j)_{j\geq 0}$, $t_j \to \infty$, such that $\|\boldsymbol{v}(\cdot,t_j)\|_{L^2(\mathbb{R})} > \lambda_p + \eta$ (for all $j \geq 0$) and $g(t) \leq \lambda_p + \eta/2$ for all $t \geq t_0$. From (2.8a), Theorem 2.2, it will then follow that

$$\|\boldsymbol{v}(\cdot,t)\|_{L^2(\mathbb{R})} > \lambda_p + \eta, \quad \forall t \geq t_0. \tag{3.9}$$

In fact, suppose that (3.9) were false, so that we had $\|\boldsymbol{v}(\cdot,\tilde{t})\|_{L^2(\mathbb{R})} \leq \lambda_p + \eta$ for some $\tilde{t} > t_0$. Taking $j \gg 1$ with $t_j > \tilde{t}$, we could then find $\hat{t} \in [\tilde{t}, t_j[$ such that $\|\boldsymbol{v}(\cdot,t)\|_{L^2(\mathbb{R})} > \lambda_p + \eta$ for all $t \in]\hat{t}, t_j]$, while $\|\boldsymbol{v}(\cdot,\hat{t})\|_{L^2(\mathbb{R})} = \lambda_p + \eta$, and so there would exist $t_* \in [\hat{t}, t_j] \setminus E_{2p}$ with $d/dt\|\boldsymbol{v}(\cdot,t)\|^2_{L^2(\mathbb{R})}$ positive at $t = t_*$. By (2.8a), we would have $\|\boldsymbol{v}(\cdot,t_*)\|_{L^2(\mathbb{R})} \leq g(t_*) \leq \lambda_p + \eta/2$, contradicting the fact that $\|\boldsymbol{v}(\cdot,t)\|_{L^2(\mathbb{R})} \geq \lambda_p + \eta$ everywhere on $[\hat{t}, t_j]$. Thus, we conclude that (3.9) cannot be false, as claimed. We then obtain, from (2.6), (3.4), (3.9):

$$\begin{aligned}\big\|\boldsymbol{v}(\cdot,t)\big\|^6_{L^2(\mathbb{R})} &\leq C_2^6 \big\|\boldsymbol{v}(\cdot,t)\big\|^4_{L^1(\mathbb{R})} \big\|\boldsymbol{v}_x(\cdot,t)\big\|^2_{L^2(\mathbb{R})} \\ &\leq g(t)^6 + \frac{2p}{2p-1}\frac{1}{\mu(t)}\big\|\boldsymbol{v}(\cdot,t)\big\|^4_{L^1(\mathbb{R})}\left(-\frac{d}{dt}\big\|\boldsymbol{v}(\cdot,t)\big\|^2_{L^2(\mathbb{R})}\right)\end{aligned}$$

for all $t \in [t_0, \infty[\setminus E_{2p}$. Recalling that $\|\boldsymbol{v}(\cdot,t)\|_{L^2(\mathbb{R})} > \lambda_p + \eta$, $g(t) \leq \lambda_p + \eta/2$, $\forall t \geq t_0$, this gives

$$-\frac{d}{dt}\big\|\boldsymbol{v}(\cdot,t)\big\|^2_{L^2(\mathbb{R})} \geq K(\eta)\cdot\mu(t), \quad \forall t \in [t_0, \infty[\setminus E_{2p}$$

for some constant $K(\eta) > 0$ independent of t, and a result like (3.6) once again follows. Since this contradicts the hypothesis (3.1), we conclude that (3.8) must be true. Observing that (3.8) is equivalent to (3.7), the proof is now complete. □

We are finally in a good position to derive Theorem B. Combining the bounds (2.17) and (3.3) above, we obtain

$$\limsup_{t\to\infty}\big\|\boldsymbol{u}(\cdot,t)\big\|_{L^\infty(\mathbb{R})} \leq \big(2mp^2\big)^{\frac{1}{p}} \cdot \mathcal{B}_\mu^{\frac{1}{p}} \cdot \mathcal{U}_p \tag{3.10}$$

for each $p \geq p_0$. This is (1.8b), except for the larger constant. We can get the smaller value given in (1.8b) using the following argument. Applying (3.7) with $q = 2p, 4p, \ldots, 2^k p$, we get

$$\mathcal{U}_{2^k p} \leq K(k; p, m)^{\frac{1}{p}} \cdot \mathcal{B}_\mu^{\frac{1}{p}(1-2^{-k})} \cdot \mathcal{U}_p, \tag{3.11a}$$

where

$$K(k; p, m) = \prod_{j=1}^{k}\big(2^{j-1}p\sqrt{m}C_2^3\big)^{2^{-j}}, \quad k \geq 1. \tag{3.11b}$$

Hence,

$$\begin{aligned}\limsup_{t\to\infty}\big\|\boldsymbol{u}(\cdot,t)\big\|_{L^\infty(\mathbb{R})} &\leq \big(2^{2k+1}mp^2\big)^{\frac{1}{2^k p}} \cdot \mathcal{B}_\mu^{\frac{1}{2^k p}} \cdot \mathcal{U}_{2^k p} \\ &\leq \Big\{\big(2^{2k+1}mp^2\big)^{2^{-k}} \cdot K(k; p, m)\Big\}^{\frac{1}{p}} \cdot \mathcal{B}_\mu^{\frac{1}{p}} \cdot \mathcal{U}_p\end{aligned}$$

for all $k \geq 1$, using (3.10) above. Letting $k \to \infty$, the estimate (1.8b) is finally obtained, observing that $\sum_{j=1}^{\infty} j2^{-j} = 2$. This completes the derivation of Theorem B:

Theorem 3.3. *Let $\mathcal{B}_\mu < \infty$. If (3.1) holds, then*

$$\limsup_{t\to\infty} \big\| \boldsymbol{u}(\cdot,t) \big\|_{L^\infty(\mathbb{R})} \le \left(\frac{3\sqrt{3}}{2\pi} \sqrt{m}\, p \right)^{\frac{1}{p}} \cdot \mathcal{B}_\mu^{\frac{1}{p}} \cdot \limsup_{t\to\infty} \big\| \boldsymbol{u}(\cdot,t) \big\|_{L^p(\mathbb{R})} \tag{3.12}$$

for every $p_0 \le p < \infty$ such that $\mathcal{U}_p \equiv \limsup_{t\to\infty} \|\boldsymbol{u}(\cdot,t)\|_{L^p(\mathbb{R})} < \infty$.

We notice in closing that, with an additional work, the analysis presented in this paper can be extended to other parabolic problems of interest. In particular, for the corresponding problems in $\mathbb{R}^n$, as, for example, the n-dimensional systems of conservation laws

$$\boldsymbol{u}_t + \sum_{j=1}^{n} \big(A_j(x,t)\boldsymbol{u} \big)_{x_j} = \mu(t)\Delta \boldsymbol{u}, \quad \boldsymbol{u}(\cdot,0) \in L^{p_0}\big(\mathbb{R}^n\big) \cap L^\infty\big(\mathbb{R}^n\big), \tag{3.13}$$

where $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$, $t \ge 0$, $\boldsymbol{u}(x,t) = (u_1(x,t), \ldots, u_m(x,t))^T \in \mathbb{R}^m$, and where $A_1(x,t), \ldots, A_n(x,t)$ are given $m \times m$ matrices with $\|A_j(x,t)\| \le B_j(T)$ on each finite strip $S_T = \mathbb{R}^n \times [0,T]$, $0 < T < \infty$, for every $1 \le j \le n$, with $\mu \in C^0([0,\infty[)$ positive, the estimates

$$\big\| \boldsymbol{u}(\cdot,t) \big\|_{L^\infty(\mathbb{R}^n)} \le K_1(m,n,p) \cdot \max\big\{ \big\| \boldsymbol{u}(\cdot,0) \big\|_{L^\infty(\mathbb{R}^n)}; \mathbb{B}_\mu(0;t)^{\frac{n}{p}} \mathbb{U}_p(0;t) \big\} \tag{3.14}$$

for arbitrary $t \ge 0$, and

$$\limsup_{t\to\infty} \big\| \boldsymbol{u}(\cdot,t) \big\|_{L^\infty(\mathbb{R}^n)} \le K_2(m,n,p) \cdot \mathcal{B}_\mu^{\frac{n}{p}} \cdot \limsup_{t\to\infty} \big\| \boldsymbol{u}(\cdot,t) \big\|_{L^p(\mathbb{R}^n)} \tag{3.15}$$

can be similarly obtained if (3.1) is satisfied, where $K_1(m,n,p) > 0$, $K_2(m,n,p) > 0$ are absolute constants depending only on m, n, p, and $\mathbb{B}_\mu(0;t)$, $\mathcal{B}_\mu$, $\mathbb{U}_p(0;t)$ are defined as in (1.4), (1.5) above, with $B(t) = (B_1(t)^2 + B_2(t)^2 + \ldots + B_n(t)^2)^{1/2}$. We hope to address other analogous extensions to more complicated parabolic systems in the future.

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